Experiment No. 6

Environment: Microsoft Windows
Tools/ Language: Oracle

OBJECTIVE: To implement the concept of sub queries & set theoretic functions.

Theory:

SET THEORETIC FUNCTIONS

The SQL UNION Operator
The UNION operator is used to combine the result-set of two or more SELECT statements.

Notice that each SELECT statement within the UNION must have the same number of columns. The columns must also have similar data types. Also, the columns in each SELECT statement must be in the same order.

SQL UNION Syntax:

```
SELECT column_name(s) FROM table1
UNION
SELECT column_name(s) FROM table2;
```

Note: The UNION operator selects only distinct values by default. To allow duplicate values, use the ALL keyword with UNION.

SQL UNION ALL Syntax:

```
SELECT column_name(s) FROM table1
UNION ALL
SELECT column_name(s) FROM table2;
```

PS: The column names in the result-set of a UNION are usually equal to the column names in the first SELECT statement in the UNION.

SQL INTERSECT Operator
The following statement combines the results with the INTERSECT operator, which returns only those rows returned by both queries:

```
SELECT column_name(s) FROM table1
INTERSECT
SELECT column_name(s) FROM table2;
```

SQL Minus Operator
The following statement combines results with the MINUS operator, which returns only unique rows returned by the first query but not by the second:

```
SELECT column_name(s) FROM table1
MINUS
SELECT column_name(s) FROM table2;
```

Sub Queries :--> A sub query is a form of an SQL statement. It is also term as nested query. The statement containing a sub query is called a parent. The parent statement uses the row return by the sub query.

It can also use for following:-

- To insert records in a target tables.
- To create tables and insert records in the table created.
- To update records in a target table.

- To create views.
- To provide values for condition in WHERE, HAVING, IN and so on used with SELECT, UPDATE and DELETE statement.

Examples:

Q: Find the students name and sid whose GPA is equal to GPA of Amy & Craig?

A: select sID, sNamefrom student
where GPA IN(select GPA from student
wheresName in('Amy', 'Craig'));

OUTPUT:

SID	SNAME
654	Amy
876	Irene
456	Doris
123	Amy
345	Craig
543	Craig
789	Gary

Q: Find the students name and sid whose GPA is equal to GPA of Amy & Craig but result should not include Amy and Craig?

A: Select sID, sName
From Student
Where GPA IN (Select GPA
From Student
Where sName IN ('Amy', 'Craig'))
And sName NOT IN ('Amy','Craig');

Q: Find the name of student with their GPA and Sid whose high school size is not equal to high school size of Doris?

A: selectsID, sName, GPA
from student
wheresizeHS NOT IN(select sizeHS
from student
wheresName='Doris');

OUTPUT: (Write any four only)

SID	SNAME	GPA
234	Bob 3.6	
345	Craig	3.5
567	Edward	2.9
678	Fay	3.8
789	Gary	3.4
987	Helen	3.7
876	Irene	3.9
765	Jay	2.9
543	Craig	3.4

Q: Find the name and Sid of student who apply to same college where Irene has applied ?

A: select sID, sNamefrom student
wheresID in(select sIDfrom apply whereCname in(select Cname
from apply wheresIDin(select sIDfrom studentwheresName='Irene')));

OUTPUT:

```
     SID   SNAME
---------- ----------
       765   Jay
       876   Irene
       987   Helen
       678   Fay
       123   Amy
       543  Craig
       345  Craig
```

Q: Give the names of colleges who receive same number of applications or more than Stanford had received?

A: select cNamefrom apply group by cName
having count(sID)>=(select count(sID)from apply
wherecName='Stanford')

OUTPUT:

```
CNAME
----------
Cornell
Stanford
```

Q: Create a table having two columns college name and number of applications they receive?

A:

Createtable collegeinfo
(cNamevarchar(10), no_of_app number(5));

insert into collegeinfo
selectcName, count(sID)from apply
group by cName;

Q: Create table and load data from already created table

Create a Table having Student id, Student Name, GPA, College Name they applied to and decision i.e.

sI D	sNam e	GP A	cNam e	decisio n

A: CreatetableApplicationDetail
AS(SELECTS.sID,sName, GPA,cName, decision
FROM STUDENT S, Apply A where S.sID=A.sID);

Q: Update Decision to N for all students who applied to same major as sID 876 have applied.

A: update APPLY
set decision='N'
WHERE major IN (SELECT major
FROM APPLY
WHEREsID = 876);

Q: Delete the Application of student who have applied to same college as of 'Irene'

A: deletefrom Apply
wherecName IN (SelectcName from Apply
wheresID IN (SelectsID from Student
wheresName = 'Irene'));

Q:Find IDs and names of student who share same name with another student.

A: Select *
From Student S1
Where exists (Select * From student S2
Where S1.sName=S2.sName)

Running above query will give WRONG result as every student of outer query will match (or each S1 student having same name as S2), we need ensure we are talking about **different person**. So,

```
Select *
From Student S1
Where exists ( Select * From student S2
             Where S1.sName=S2.sName
andS1.sID <> S2.sID)
```

Q: Find IDs of students that applied to all colleges.

Insert these to get student who applied to all college.

```
insert into Apply values (123, 'MIT', 'CSE', 'N');
insert into Apply values (123, 'Harvard', 'CSE', 'N');
```

A: Select s.sID, s.sName
From Student S
Where **NOT Exists** ((Select C.cName from College C)
MINUS
(Select A.cName from Apply A where A.sID=S.sID));

Kindly delete the above two rows

```
Delete from apply where
major='CSE';
```

Practical Assignment - 6

Department: Computer Engineering & Applications **Course:** BCA **Subject:** Database Management System Lab (BCAC0816) **Year:** 1ST **Semester:** 2ND	ऋते ज्ञानान्न मुक्तिः

Create these tables which consist of following attributes

College

Column Name	Data type	Size
cName	varchar2	10
cstate	varchar2	10
enrollment	int	

Student

Column Name	Data type	Size
sID	int	
sName	varchar2	10
CGPA	number	2,1
marks	int	
DoB	date	

Apply

Column Name	Data type	Size
sID	int	
cName	varchar2	10
major	varchar2	20
decision	char	1

Insert the following data to these tables:

Student

sID	sName	CGPA	Marks
123	Amit	3.9	1000
234	Balbir	3.6	1500
345	chirag	3.5	500
456	Dev	3.9	1000
567	Eshan	2.9	2000
678	Faizal	3.8	200
789	Garvit	3.4	800
987	Himani	3.7	800
876	Ishan	3.9	400
765	Jay	2.9	1500
654	Amit	3.9	1000
543	chirag	3.4	2000

College

cName	state	enrollment
Stanford	CA	15000
Berkeley	CA	36000
MIT	MA	10000
Cornell	NY	21000
Harvard	MA	50040

Apply

sID	cName	major	decision
123	Stanford	CS	Y
123	Stanford	EE	N
123	Berkeley	CS	Y
123	Cornell	EE	Y
234	Berkeley	biology	N
345	MIT	bioengineering	Y
345	Cornell	bioengineering	N
345	Cornell	CS	Y
345	Cornell	EE	N
678	Stanford	history	Y
987	Stanford	CS	Y
987	Berkeley	CS	Y
876	Stanford	CS	N
876	MIT	biology	Y
876	MIT	marine biology	N
765	Stanford	history	Y
765	Cornell	history	N
765	Cornell	psychology	Y
543	MIT	CS	N

State the SQL Queries for each of the following:

1. IDs and names of students who have applied to major in CS at some college.
2. Find ID and name of student having same marks as *Dev*.
3. Find ID and name of student having same marks as *Dev* but result should not include *Dev*.
4. Find the name of student with their CGPA and Sid whose CGPA not equal to CGPA of Ishan?
5. Find college where any student having their name started from J have applied?
6. Find all different major where *Ishan* has applied?
7. Find IDs of student and major who applied in any of major *Ishan* had applied to?
8. Find IDs of student and major who applied in any of major *Ishan* had applied to? But this time exclude *Ishan* ID from the list.
9. Give the number of colleges *Jay* applied to? (*Remember count each college once no matter if he applied to same college twice with different major*)
10. Find sID of student who applied to more or same number of college where *Jay* has applied?
11. Find the college with highest enrollment.
12. Find name of student having lowest CGPA.
13. Find the most popular major.